Lab 5 - Segmentation & 3D Slicer

1. From classical segmentation to Deep Learning
2. U-Net
3. Dice for training and evaluation
4. 3D Slicer for annotation



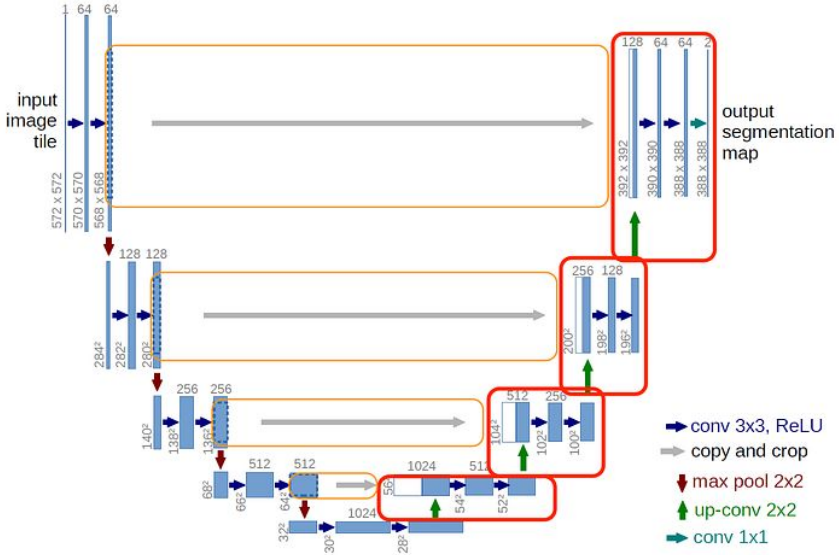


1. From classical segmentation to Deep Learning

- **Segmentation** task = **one label per pixel** (0 = background, 1 = LV).
- Useful for downstream tasks (e.g., ejection fraction estimation).

	Classical	Deep Learning
What drives the decision?	**Rules** chosen by an expert (intensity, edges, smoothness)	**Patterns** learned from manually traced examples
Input needed per image	**Seed** point (region growing) or **tuned** parameters (thresholding)	**Nothing**: image in, mask out
Where the effort goes	**Designing** and **re-tuning** the method for each dataset	**Annotating** enough training images
Typical failures	**Sensitive** to speckle and dropout if **not properly handled**	**Domain shift**: plausible-looking but wrong masks

2. U-Net (Ronneberger et al., MICCAI 2015)



- Encoder: 2 × (3×3 conv + ReLU) + 2×2 max-pool. Resolution halves, channels double → **what**.
- Decoder: 2×2 up-conv + concatenate the skip + 2 × (3×3 conv + ReLU) → **where**.
- **Skip** connections: restore fine detail
- Output: 1×1 conv + sigmoid/ softmax.
- U-Net variants were among the best on CAMUS.

Image from https://www.comet.com/site/blog/deep-learning-for-image-segmentation-u-net-architecture/

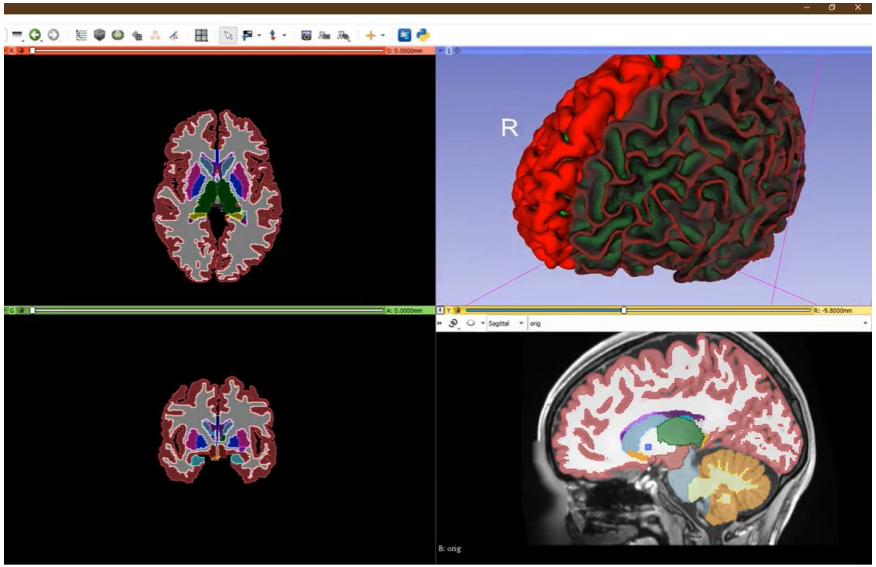
3. Dice for training and evaluation

- **Idea**: **how much** do the predicted mask and the ground truth **overlap**?
- 1 = perfect, 0 = none. It ignores true negatives (= **robust to class imbalance)**.

	Evaluation (after train)	Training (loss)
Prediction use	**Binary mask**: $\hat{y}_i = 1$ if $p_i > 0.5$, else 0	**Probabilities**: $p_i \in [0, 1]$
Formula	$\mathrm{DSC} = \frac{2\,\lvert A \cap B \rvert}{\lvert A \rvert + \lvert B \rvert} = \frac{2\,\mathrm{TP}}{2\,\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}$	$\mathcal{L}_{\mathrm{Dice}} = 1 - \frac{2\sum_i p_i g_i + \varepsilon}{\sum_i p_i + \sum_i g_i + \varepsilon}$
Why this version	Compares the **final mask with** the **ground truth**	Thresholding has zero gradient; **probabilities give a learning signal**

4. 3D Slicer for annotation (https://slicer.org/)

- **Free**, **open-source** platform for medical image **visualization** and **analysis**.
- **Modular** and pipeline-friendly: reads DICOM / NIfTI and exports labelmaps.
- **Manual** to **AI-assisted** segmentation: fully manual, semi-automated, or AI-enabled.



- Segment **Editor effects**: Draw, Level tracing, Thresholding, Islands, Smoothing.
- Use-case: the in-house dataset has no ground truth. This could be used to build a **small test** with the **segmented LV** (binary mask) set on which to **evaluate any method** (classical or a DL model fine-tuned on a different dataset) with Dice.